SCIENTIFIC REASONING

Theoretical Foundations, Measurement Models, and Validation Studies

Rossella Caliciuri
Supervision: Margherita Lanz

TWO GUIDING PILLARS

1 THEORETICAL

SCIENTIFIC REASONING

WHAT IS SCIENTIFIC REASONING? → WHY DOES IT MATTER TODAY? → HOW HAS IT BEEN MEASURED?

2 METHODOLOGICAL

VALIDITY AND VALIDATION IN PSYCHOLOGICAL MEASUREMENT

- VALIDITY IS NOT A PROPERTY OF A TEST, BUT OF THE INFERENCES DRAWN FROM TEST SCORES
- VALIDATION REQUIRES MULTIPLE, COMPLEMENTARY SOURCES OF EVIDENCE
- WEAK VALIDATION UNDERMINES CUMULATIVE SCIENCE

TO CLARIFY THE **CONCEPTUAL FOUNDATIONS** OF SCIENTIFIC REASONING, WHILE PROVIDING **CUMULATIVE VALIDITY EVIDENCE** FOR ITS MEASUREMENT

TO ACHIEVE THIS GOAL, THREE EMPIRICAL STUDIES

OSF

JANE LOEVINGER'S TRIPARTITE FRAMEWORK

SUBSTANTIVE	STRUCTURAL	EXTERNAL
How is the construct conceptually defined?	What are the psychometric properties?	How does it relate to other constructs or outcomes?
multiple competing definitions, reflecting richness but **lacking a shared theoretical framework**	many instruments exist, but their psychometric properties and construct relations are **inconsistently reported**, and **none were developed using a coherent validity framework**	

1 Umbrella Review

to systematically synthesize how scientific reasoning has been conceptualized, measured, and validated across existing systematic reviews and meta-analyses

2 Validation Studies

to adapt and provide validity evidence for a Scientific Reasoning Scale in Italy

- clear **theoretical grounding**: Scientific Reasoning Scale
- careful **adaptation of items**: backtranslation, cognitive and expert interviews

psychometric evaluation within CTT
- **Pilot** testing (N = 337)
- **CFA** to test the hypothesized factor structure (N = 897)
- **ω** to estimate internal consistency
- **Measurement and structural invariance** analyses

SEM testing relation with theoretically relevant variables and outcomes

3 Accumulating Validity Evidence

to refine the scale's structural validity using IRT and a person-centered lens (LCA) to gain a deeper understanding of item functioning beyond CTT

psychometric evaluation within IRT
- **2PL**
- **EAP** scores for person ability estimation
- **TIF** to assess measurement precision

exploratory psychometric evaluation within LCA
- **Misclassified cases**
- **Classification error**
- **KL Distance**
- **Item Endorsment Odds Ratio**
- **Class Homogeneity**

ADVANCING THE THEORY OF SCIENTIFIC REASONING

- Frames scientific reasoning as an integrated competence that combines cognitive, epistemic, and civic aspects.
- Provides a validated and culturally adapted instrument for its assessment.

ADVANCING THE THEORY OF VALIDITY

- Demonstrates that validation is a cumulative process, not a single event.
- Integrates multiple frameworks (CTT, IRT, LCA) to strengthen validity evidence.

NO VALIDATION PROCESS IS EVER COMPLETE:
EACH STEP HELPS CLARIFY THE CONSTRUCT WHILE SIMULTANEOUSLY RAISING NEW QUESTIONS

Main future directions:
- improving reliability and content coverage
- exploring new theoretical and methodological horizons
- enhancing generalization

rossella.caliciuri@unicatt.it

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